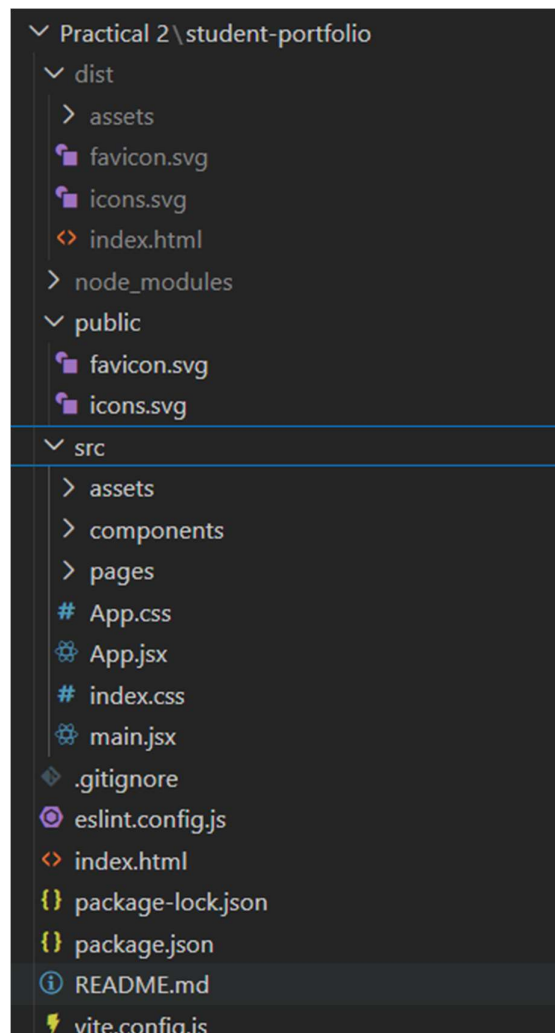


Practical 2 : State Management and Routing in React

Problem Definition :

- Extend the portfolio application from Practical 1 by adding React Router.
- Implement a navigation bar with at least 3 routes: Home, Projects, and Contact.
- Each route must render a distinct component without a full page reload.
- Add at least 2 useState variables used meaningfully one for toggling UI visibility and one for managing a form input on the Contact page.
- The Contact page must include a controlled input that captures and displays user input in real time.

Folder Structure:



Code:**Index.html:**

```
<!doctype html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <link rel="icon" type="image/svg+xml" href="/favicon.svg" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>student-portfolio</title>
    <script type="module" crossorigin src="/assets/index-wjLdS5mD.js"></script>
    <link rel="stylesheet" crossorigin href="/assets/index-ClJC4mhn.css">
  </head>
  <body>
    <div id="root"></div>
  </body>
</html>
```

About.jsx:

```
function About() {
  return (
    <section className="section">
      <h2>About Me</h2>
      <p>
        my name is rudra and i am a student.
      </p>
      <div className="details">
        <p><strong>College :</strong> DEPSTAR </p>
        <p><strong>Subject :</strong> ITUE301 - ADVANCED WEB DEVELOPMENT
FRAMEWORKS</p>
        <p><strong>Semester :</strong>5th Sem</p>
        <p><strong>Enrollment No :</strong> D25DIT081</p>
        <p><strong>Division :</strong> 5IT </p>
      </div>
    </section>
  );
}

export default About;
```

Contact.jsx

```
import { useState } from "react";
```

```

function Contact() {
  const [message, setMessage] = useState("");

  return (
    <section className="section">
      <h2>Contact Page</h2>
      <p>Send a message below and see real-time updates:</p>

      <div className="details" style={{ marginTop: "25px" }}>
        <label htmlFor="messageInput" style={{ display: "block", marginBottom: "10px", color:
"#bdbdbd" }}>
          Write Message:
        </label>
        <input
          id="messageInput"
          type="text"
          value={message}
          onChange={(e) => setMessage(e.target.value)}
          placeholder="Type your message"
          style={{
            padding: "12px 20px",
            width: "100%",
            maxWidth: "450px",
            borderRadius: "8px",
            border: "1px solid #333",
            background: "#1b1b1b",
            color: "#fff",
            fontSize: "16px",
            outline: "none"
          }}
        />

        <div style={{ marginTop: "25px" }}>
          <p><strong>Entered Text (Real Time):</strong></p>
          <p style={{ color: "#fff", marginTop: "10px" }}>
            {message || "No text entered yet..."}
          </p>

          <div style={{ marginTop: "15px" }}>
            <p><strong>Character Count:</strong> {message.length}</p>
          </div>
        </div>
      </section>
    );
  }
}

export default Contact;

```

Footer.jsx:

```
function Footer(props) {  
  return (  
    <section className="footer">  
      <h2>Contact</h2>  
      <p>Email : {props.email}</p>  
      <p>Phone : {props.phone}</p>  
      <p>City : {props.city}</p>  
      <hr />  
      <p>© 2026 Student Portfolio</p>  
    </section>  
  );  
}  
  
export default Footer;
```

Header.jsx:

```
function Header(props) {  
  return (  
    <section className="hero">  
      <h1 style={{ color: props.themeColor }}>{props.title}</h1>  
      <h2>{props.name}</h2>  
      <p>{props.role}</p>  
      <button>View Skills</button>  
    </section>  
  );  
}
```

```
export default Header;
```

Home.jsx

```
import { useState } from "react";
import Header from "./Header";
import About from "./About";
import Skills from "./Skills";

function Home() {
  const [showAbout, setShowAbout] = useState(true);

  const skills = [
    "HTML",
    "CSS",
    "JavaScript",
    "React",
    "Node js",
    "Vite"
  ];

  return (
    <div>
      <Header
        title="Student Portfolio"
        name="Rudra Maiyariya"
        role="Computer Engineering Student"
        themeColor="#ffffff"
      />

      <div style={{ textAlign: "center", margin: "10px 0" }}>
        <button onClick={() => setShowAbout(!showAbout)}>
          {showAbout ? "Hide About Section" : "Show About Section"}
        </button>
      </div>

      {showAbout && <About />}

      <Skills skillList={skills} />
    </div>
  );
}

export default Home;
```

Navbar.jsx:

```
import { NavLink } from "react-router-dom";
```

```

function Navbar() {
  return (
    <nav className="navbar">

      <h2 className="logo">Portfolio</h2>

      <ul>

        <li>
          <NavLink to="/" className={{ { isActive }} => (isActive ? "active" : "")}>
            Home
          </NavLink>
        </li>

        <li>
          <NavLink to="/projects" className={{ { isActive }} => (isActive ? "active" : "")}>
            Projects
          </NavLink>
        </li>

        <li>
          <NavLink to="/contact" className={{ { isActive }} => (isActive ? "active" : "")}>
            Contact
          </NavLink>
        </li>

      </ul>

    </nav>
  );
}

export default Navbar;

```

Project.jsx:

```

function Projects() {
  return (
    <section className="section">
      <h2>My Projects</h2>
      <div className="details">
        <div style={{ marginBottom: "25px" }}>
          <h3 style={{ fontSize: "22px", marginBottom: "8px" }}>Practical 1 - Portfolio
Foundation</h3>
          <p>Built the initial React portfolio application with reusable components (Header, About,
Skills, Footer) and modern UI styling.</p>
        </div>
        <div style={{ marginBottom: "25px" }}>

```

```
        <h3 style={{ fontSize: "22px", marginBottom: "8px" }}>Practical 2 - React Router &  
Hooks</h3>  
        <p>Extended the portfolio application with multi-page client-side routing using React  
Router and interactive state management with useState hooks.</p>  
      </div>  
    </div>  
  </section>  
);  
}  
  
export default Projects;
```

Skill.jsx:

```
function Skills(props) {  
  
  return (  
  
    <section className="section">  
  
      <h2>Technical Skills</h2>  
  
      <div className="skills">  
  
        {props.skillList.map((skill) => (  
  
          <span className="skill" key={skill}>  
            {skill}  
          </span>  
  
        ))}  
  
      </div>  
  
    </section>  
  
  );  
}  
  
export default Skills;
```

NotFound.jsx:

```
function NotFound() {  
  return (  
    <section className="section">  
      <h2>404 - Page Not Found</h2>  
      <p>The page you are looking for does not exist.</p>  
    </section>  
  );  
}
```

```
    </section>
  );
}

export default NotFound;
```

App.jsx

```
import { BrowserRouter, Routes, Route } from "react-router-dom";
import "./App.css";

import Navbar from "./components/Navbar";
import Footer from "./components/Footer";
import Home from "./components/Home";
import Projects from "./components/Projects";
import Contact from "./components/Contact";
import NotFound from "./pages/NotFound";

function App() {
  return (
    <BrowserRouter>
      <div>
        <Navbar />

        <Routes>
          <Route path="/" element={<Home />} />
          <Route path="/projects" element={<Projects />} />
          <Route path="/contact" element={<Contact />} />
        </Routes>

        <Footer
          email="rudra@email.com"
          phone="+91 1234567891"
          city="Porbandar"
        />
      </div>
    </BrowserRouter>
  );
}

export default App;
```

Index.css:

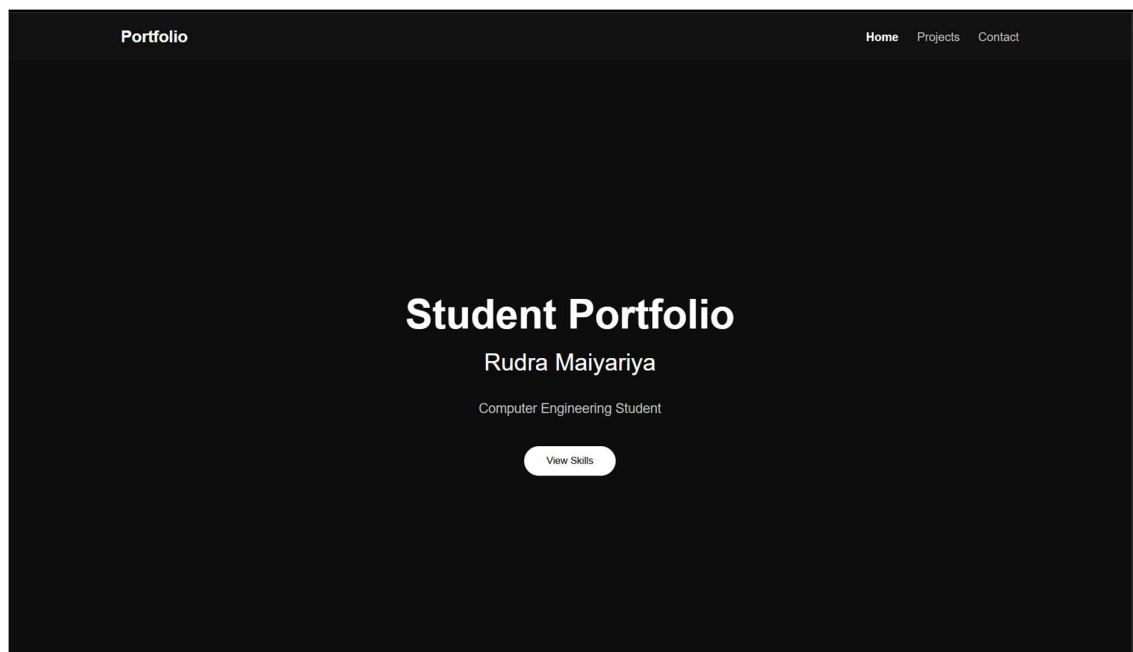
```
body{
  margin:0;
}
```

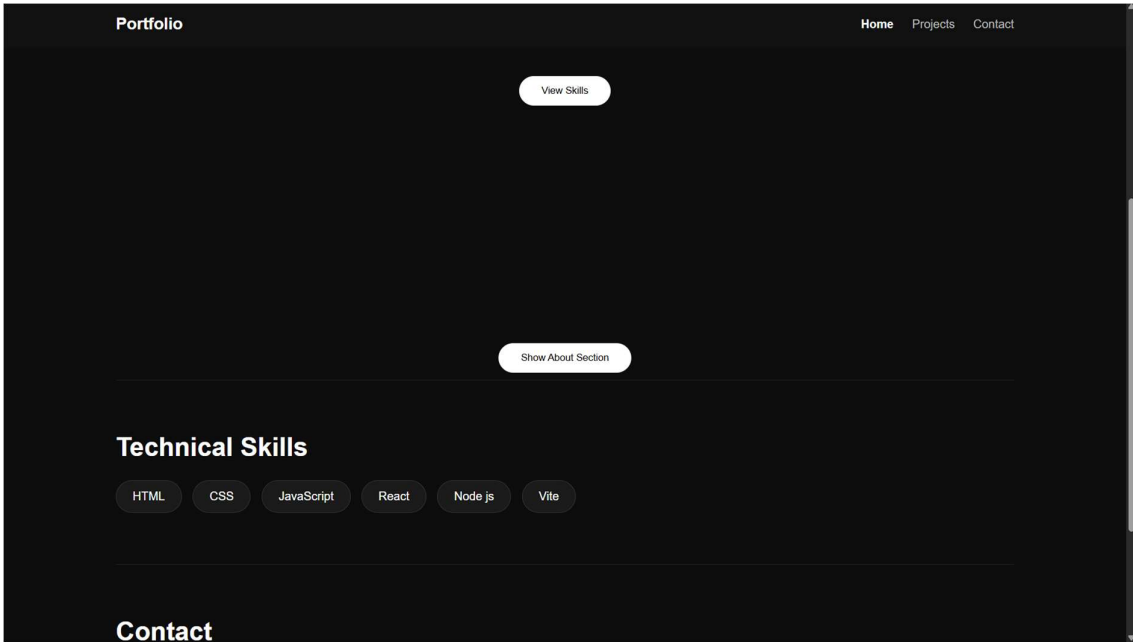
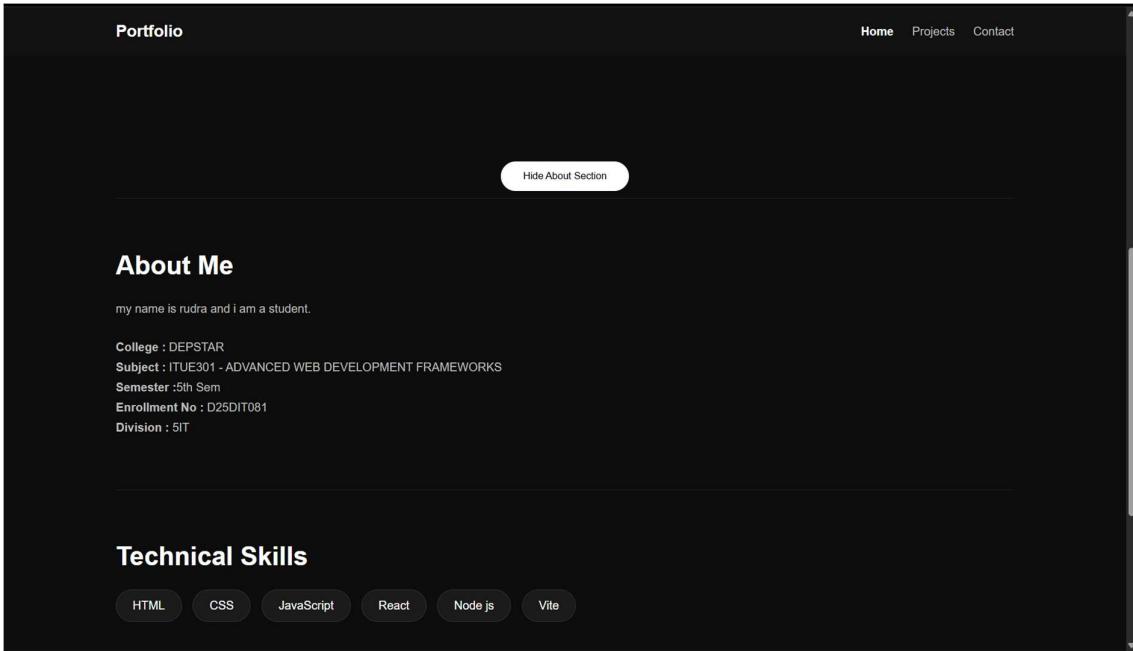
Main.jsx:

```
import { StrictMode } from 'react'
import { createRoot } from 'react-dom/client'
import './index.css'
import App from './App.jsx'

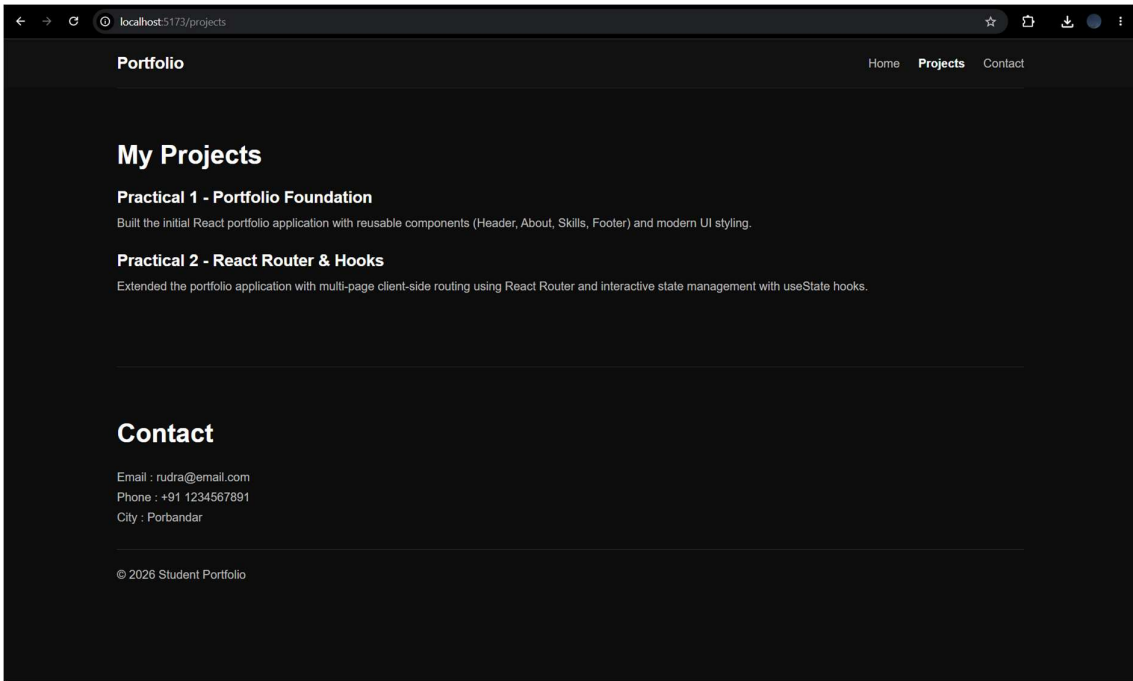
createRoot(document.getElementById('root')).render(
  <StrictMode>
    <App />
  </StrictMode>,
)
```

Output:

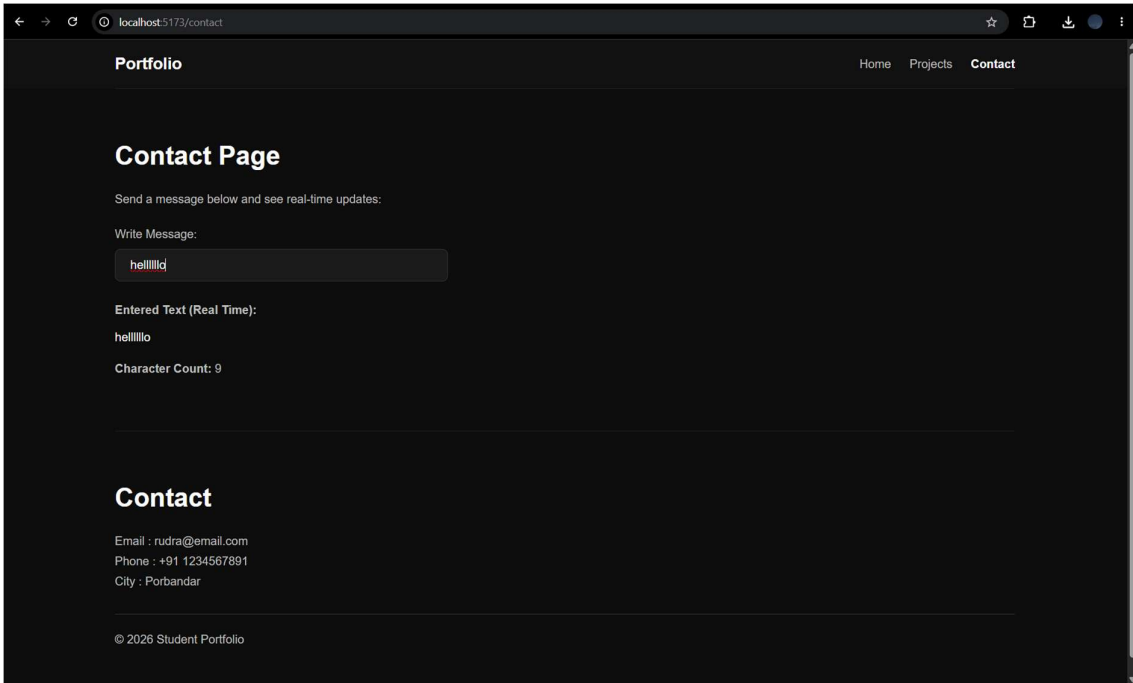




Project:



Contact:



Supplementary Problems :

- **Add a 404 Not Found route that renders a custom error component for unknown paths.**

NotFound.jsx:

```
import React from 'react';
import { Link } from 'react-router-dom';

function NotFound() {
  return (
    <div style={{ textAlign: 'center', padding: '50px' }}>
      <h2>404 - Page Not Found</h2>
      <p>Oops! The page you are looking for does not exist.</p>
      <Link to="/" style={{ color: 'blue', textDecoration: 'underline' }}>Go back to Home</Link>
    </div>
  );
}
export default NotFound;
```

App.jsx:

```
<Route path="*" element={ <NotFound /> } />
```

- **Implement a dark/light mode toggle using useState that applies a CSS class to the root element.**

App.jsx:

```
import React, { useState, useEffect } from 'react';

function App() {
  const [isDarkMode, setIsDarkMode] = useState(false);

  useEffect(() => {
    // Toggles a CSS class on the root body element
    if (isDarkMode) {
      document.body.classList.add('dark-mode');
    } else {
      document.body.classList.remove('dark-mode');
    }
  }, [isDarkMode]);

  return (
    <div>
      <button onClick={() => setIsDarkMode(!isDarkMode)}>
        Switch to {isDarkMode ? 'Light' : 'Dark'} Mode
      </button>
      { /* Rest of the app components */ }
    </div>
  );
}
```

```
    </div>
  );
}
```

- **Store the contact form input in state and display a live character count below the input.**

Contact.jsx:

```
import React, { useState } from 'react';

function Contact() {
  const [message, setMessage] = useState("");

  return (
    <section>
      <h2>Contact Me</h2>
      <textarea
        value={message}
        onChange={(e) => setMessage(e.target.value)}
        placeholder="Type your message..."
        rows="5"
        style={{ width: '100%' }}
      />
      { /* Live character count display */ }
      <p style={{ color: '#555', fontSize: '14px' }}>
        Character count: {message.length}
      </p>
    </section>
  );
}
export default Contact;
```